

Control of an Isomerization Column/Reactor Process

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ABSTRACT: The control of a distillation column with a side reactor is studied for the conversion of *n*-butane into isobutane for use in the alkylation process. Fresh feed of a mixture of propane, isobutane, *n*-butane, and isopentane is fed to a distillation column. A vapor sidestream is withdrawn in the stripping section of the column and fed to an isomerization reactor in which some of the *n*-butane is converted to isobutane. Reactor effluent is fed back into the column in the rectifying section. Isopentane is removed in the bottoms, and moderate-purity isobutane is removed in the distillate. The control of the purity of the isobutane product can be achieved in two alternative ways: manipulating sidestream flow rate to the reactor or manipulating reflux ratio. The latter method is demonstrated to be better because manipulating sidestream flow rate presents potential control problems due to the existence of multiple steady states.

1. INTRODUCTION

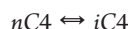
Alkylation processes of C3 and C4 olefins with isobutane were developed over six decades ago and were used during World War II for the production of high-octane aviation gasoline. Alkylation remains a vital process in many refineries to produce a very high-quality alkylate for use in gasoline blending.

The chemistry involves reacting C3 or C4 olefins produced in catalytic cracking with isobutane to produce alkylate (mostly isooctane)



Crude oil contains both isobutane and normal butane, but there is usually insufficient isobutane to meet the alkylation requirements. Therefore many refineries use isomerization of *n*-butane to provide the needed isobutane.

The isomerization reaction occurs in the gas phase and is mildly exothermic. A packed tubular reactor is used and operated adiabatically. The reaction is reversible, so per-pass conversion is limited by chemical equilibrium and favored by low temperature



This paper considers a process with a fixed equipment design (column stages and reactor size), and a control system is developed. The proposed control structure should be applicable to modified designs as long as minimum-tray limitations are avoided among the locations of the fresh feed, sidestream draw-off, column base, and reactor effluent return.

2. PROCESS STUDIED

Figure 1 shows the flowsheet of the process. A fresh feed stream of 100 lb-mol/h is fed to a large 62-stage distillation column on Stage 30. The feed composition is 1 mol % propane (C3), 20 mol % isobutane (iC4), 74 mol % *n*-butane (nC4), and 5 mol % isopentane (iC5). The heavy iC5 is removed in the bottoms at a purity of 95 mol %. The distillate specification is 8 mol % nC4 impurity. Of course, all the propane leaves in the distillate. A reflux-drum pressure of 90 psia is selected so that cooling water can be used in the overhead condenser (reflux-drum temperature is 119 °F).

Most of the nC4 in the feed stream must be converted to iC4. This is achieved in a reactor that is fed by a vapor sidestream withdrawn from the stripping section of the column (Stage 45). A vapor sidestream is used so that the iC5 impurity is smaller than would be the case if a liquid sidestream were withdrawn. The vapor is condensed, pumped up to 125 psia, vaporized, and fed into a gas-phase tubular reactor. The sidestream flow rate is 220 lb-mol/h with a composition of 8.13 mol % iC4.

Reactor inlet temperature is 165 °F, which is the dewpoint temperature of the sidestream at 125 psia. The exothermic reaction in the adiabatic reactor raises the temperature to 203.5 °F. The reactor effluent is 38.19 mol % iC4 and is essentially at chemical equilibrium at this temperature. The reactor is 1.7 ft in diameter and 17 ft in length. It is packed with a catalyst that has a solid density of 100 lb/ft³ and a bed voidage of 0.5.

The reactor effluent vapor is fed back into the column in the rectifying section (Stage 20) where the pressure is 91.9 psia. The 125 psia pressure in vaporizer is selected so that the vapor can flow back into the column through a control valve.

3. REACTION KINETICS

The isomerization reaction kinetics are assumed to be fast enough so that chemical equilibrium is attained in the reactor with a typically small vapor residence time (3.6 s). Initial steady-state simulations in Aspen Plus used an RGIBBS reactor model in which C3 and iC5 are specified to be inert and all four components are potential products.

The reaction is equimolar, so pressure should not affect chemical equilibrium. The exothermic reaction means that low temperatures increase the chemical equilibrium constant. The predictions of the RGIBBS reactor are that the conversion of a pure nC4 feed stream to iC4 when operating at 170 °F is 41.35%. When operating at 200 °F, conversion drops to 38.97%.

During the dynamic control studies using Aspen Dynamics, simulation difficulties arose with error messages indicating

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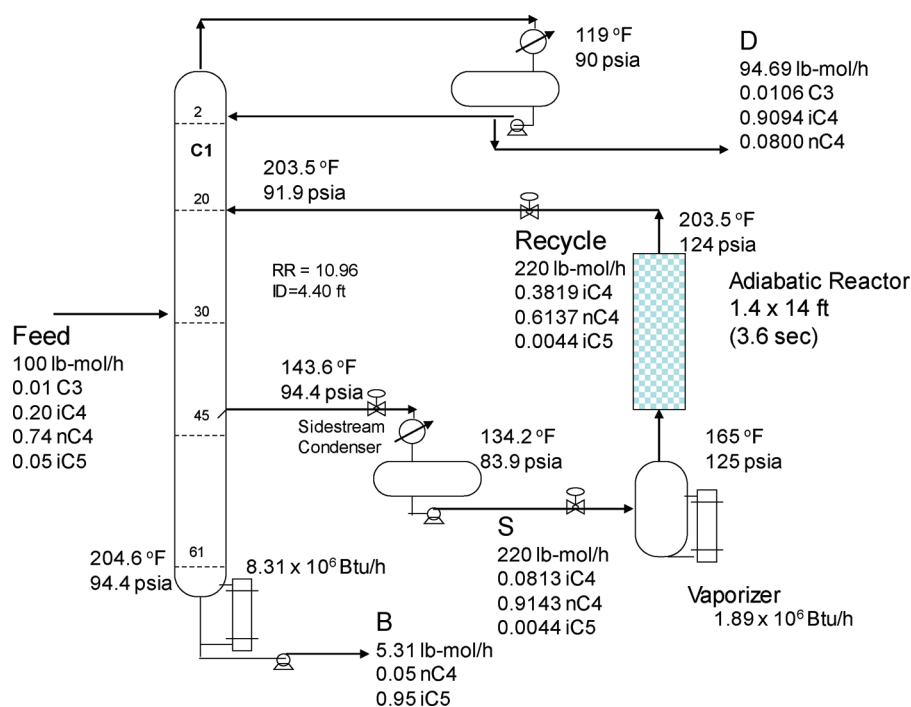


Figure 1. Isomerization flowsheet.

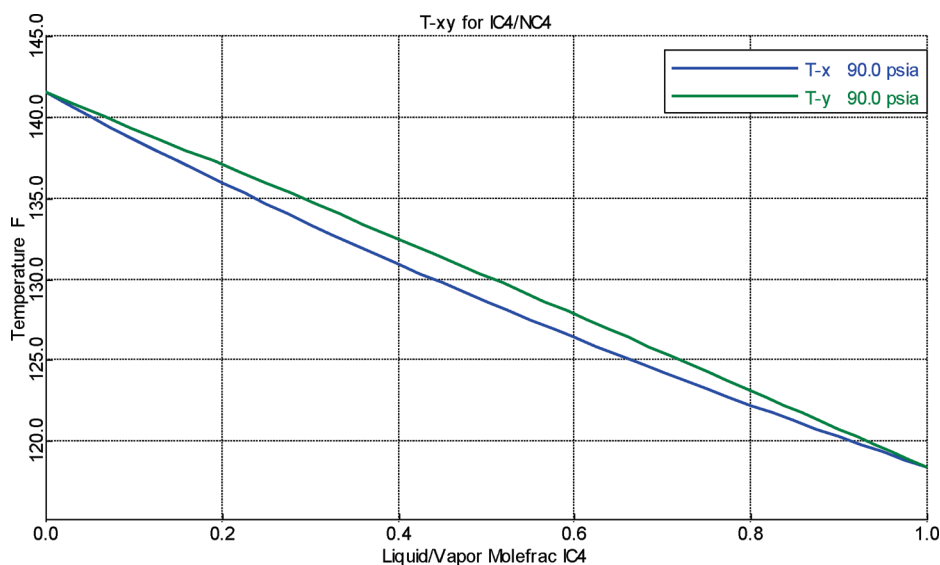


Figure 2. Txy diagram for iC4/nC4 system at 90 psiz.

nonlinear solver problems. Apparently using RGIBBS in dynamic simulations is not robust despite the fact that it can be exported into Aspen Dynamics.

As a work-around, empirical power-law kinetic reactions were developed that fit the RGIBBS predictions at the two temperatures. The forward and reverse overall reaction rates are given below with the Aspen required units of $\text{kmol s}^{-1} \text{m}^{-3}$

$$\mathcal{R}_F = P_{nC4} k_{oF} e^{-E_F/RT}$$

$$\mathcal{R}_R = P_{iC4} k_{oR} e^{-E_R/RT}$$

Partial pressures are in Pascals. An activation energy for the forward reaction was arbitrarily set at 20,000 Btu/lb-mol, and a

pre-exponential factor was set at unity. Then various combinations of activation energy and pre-exponential factor for the reverse reaction were tested until a set was found that matched the RGIBBS reactor results for conversion at the two temperatures (41.35% at 170 °F and 38.97% at 200 °F)

$$\mathcal{R}_F = P_{nC4}(1)e^{-20,000/RT}$$

$$\mathcal{R}_R = P_{iC4}(15.6)e^{-23,000/RT}$$

The reactor in the process is made large enough to provide a typical gas-phase residence time of 3.6 s. An aspect ratio of 10 is assumed, which gives a diameter of 1.7 ft and a length of 17 ft.

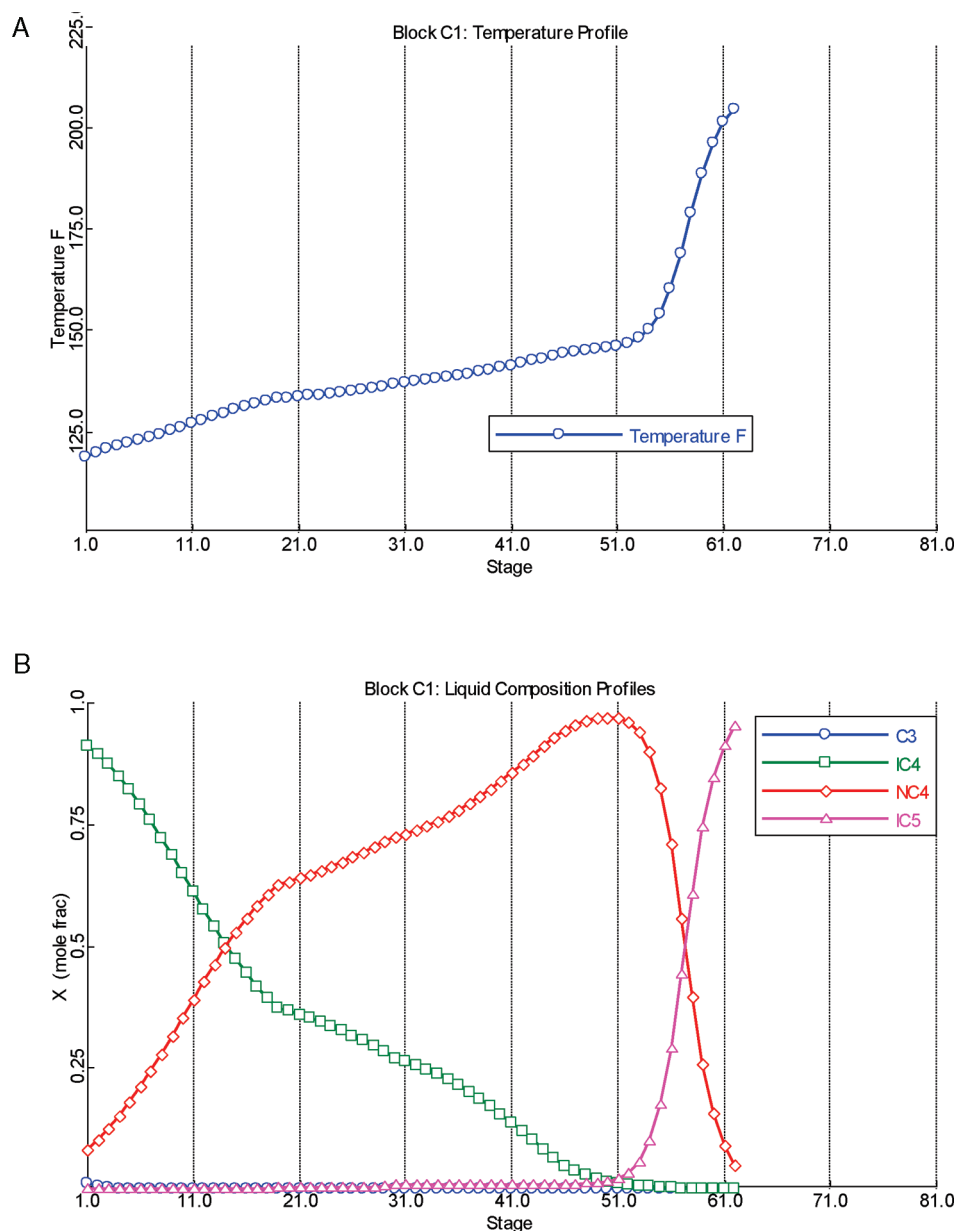


Figure 3. A. Temperature profile and B. composition profiles.

4. PHASE EQUILIBRIUM

The isobutane/n-butane separation is fairly difficult, so a column with a large number of stages (62) and a high reflux ratio (10.96) is required. Chao-Seader physical properties are used in the Aspen simulations.

Figure 2 gives the Txy diagram for the iC4/nC4 system at 90 psia, the operating pressure of the column. Figure 3 gives temperature and composition profiles in the column at steady-state conditions.

5. EFFECTS OF DESIGN OPTIMIZATION VARIABLES

There are two design optimization variables in this process once the pressure and the number of stages and feed and withdrawal stages have been set. They are reflux ratio and sidestream flow rate. The purity specifications on the bottoms (95 mol % iC5) and on the distillate (8 mol % nC4) can be

achieved with any number of combinations of reflux ratio and sidestream flow rate.

For a fixed sidestream flow rate, increasing reflux ratio increase energy requirements in the column reboiler. For a fixed reflux ratio, increasing sidestream flow rate increases energy consumption in the vaporizer. Both the reboiler (base temperature 204.6 °F) and vaporizer (temperature 165 °F) use low-pressure steam, so their energy costs per Btu are the same.

The steady-state design is established by finding the combination of reflux ratio and sidestream flow rate that gives the minimum total energy consumption (reboiler plus vaporizer). Figure 4 gives results for a range of sidestream flow rates. The lower left graph shows that the reflux ratio required to achieve the product specifications decreases as the sidestream flow rate increases. The top left graph shows that reboiler energy decreases and vaporizer energy increases as sidestream flow rate increases. Therefore a minimum occurs in the total energy consumption, as

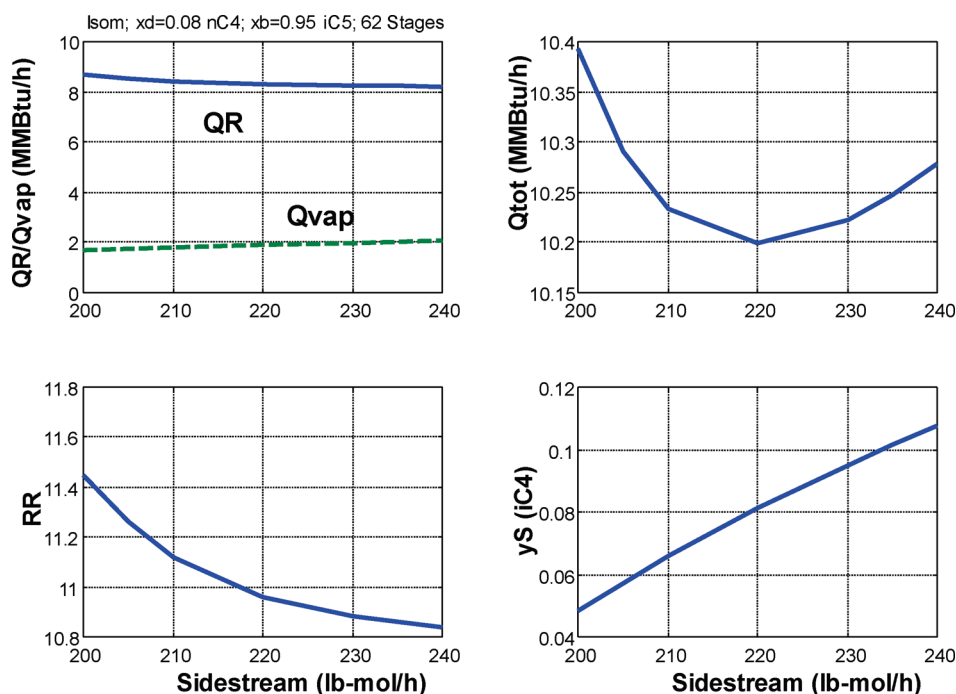


Figure 4. Effect of RR and S on energy consumption.

$x_B(\text{iC5}) = 0.95$ with B; $x_D(\text{nC4}) = 0.08$ with RR

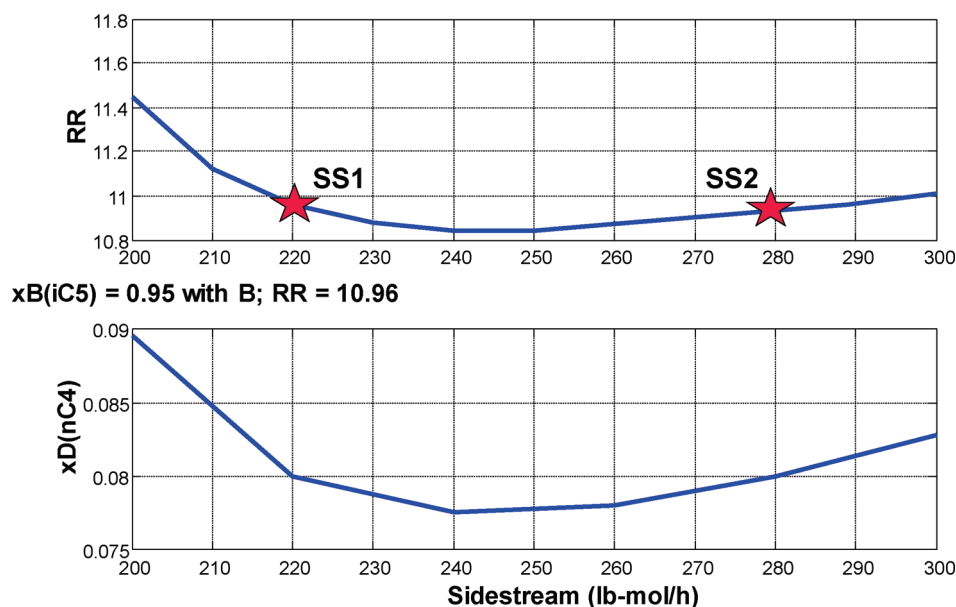


Figure 5. Multiple steady states.

shown in the upper right graph of Figure 4. A sidestream flow rate of 220 lb-mol/h is selected as the steady-state design point, which has a reflux ratio of 10.96.

Notice that the iC4 composition of the sidestream vapor (y_S) increases as sidestream flow rate increases. The reactor effluent composition changes very little from the chemical equilibrium constraint, and the number of moles of nC4 that must be converted to iC4 remains the same. So the change in composition through the reactor is smaller at larger sidestream flow rates.

6. DYNAMIC CONTROL

Once the steady-state conditions have been established, the Aspen Plus file can be exported to Aspen Dynamics after vessel sizes have been specified. The reflux drum, column base, vaporizer, and sidestream drum are all sized to provide 5 min of liquid holdup when at 50% level. The reflux drum and reboiler sizes are 6×12 ft. The vaporizer and drum are 3.5×7 ft. The size of the reactor (7×17 ft) is specified in the steady-state design. Aspen Tray Sizing is used to determine the column diameter (4.4 ft).

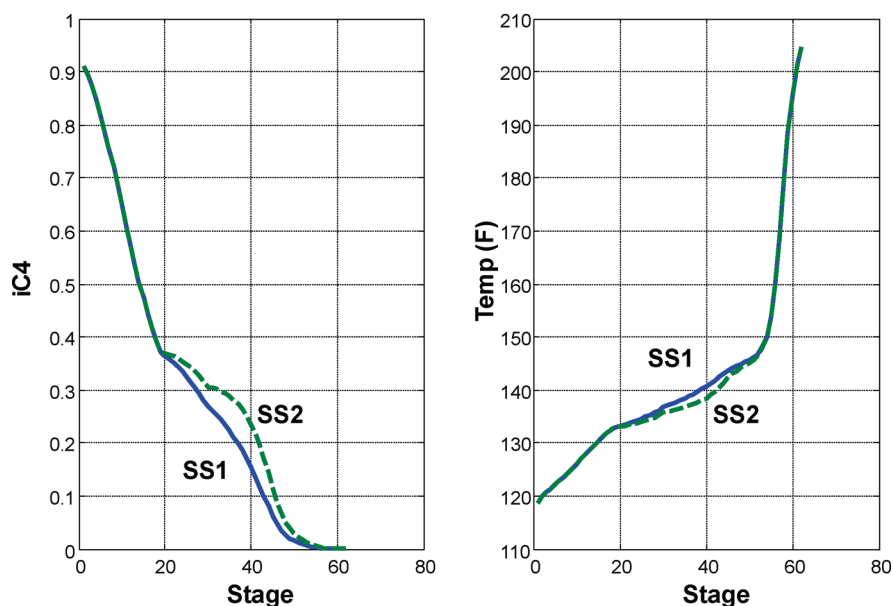
SS1/SS2; $z_{iC4}=0.2$; $RR=10.96$; $S=220/279.5$ lb-mol/h

Figure 6. Composition and temperature profiles at two steady states.

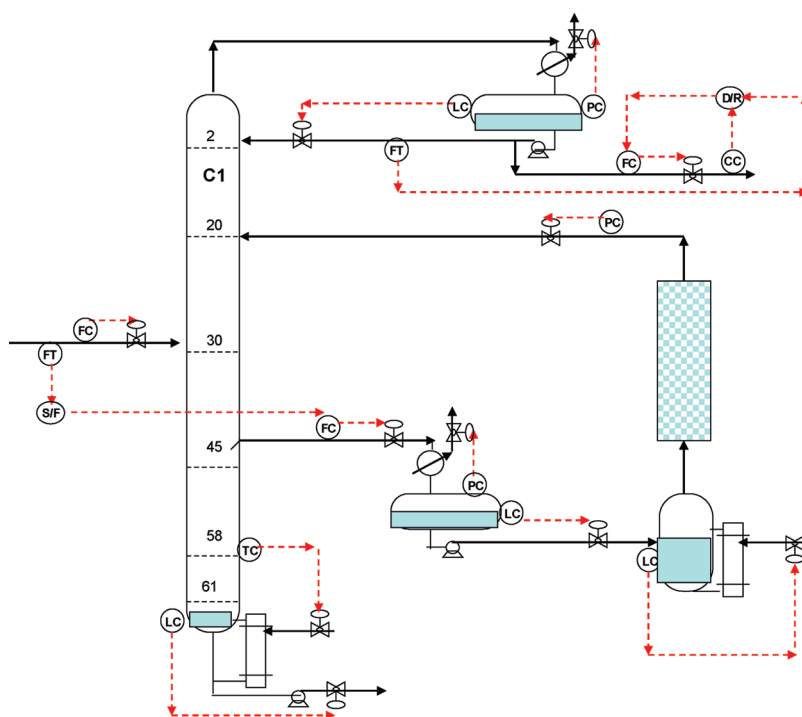


Figure 7. Isomerization control structure.

The dominant control structure question is how to control distillate purity. Should it be controlled by manipulating reflux ratio or by manipulating sidestream flow rate? Figure 5 provides an answer to this question. The upper graph is an extension of the lower left graph in Figure 4 out to higher sidestream flow rates. Both product specifications are held at their desired values. The curve is not monotonic, showing that there are two values of sidestream flow rate for the same reflux ratio that meet specifications. Thus there are two steady-state conditions at which the same reflux ratio is required. The condition labeled SS1 in Figure 5 has a sidestream

flow rate of 220 lb-mo/h. The condition labeled SS2 has a sidestream flow rate of 279.5 lb-mol/h. Multiple steady-state could lead to difficult control problems.

Figure 6 shows the temperature and composition profiles at these two steady-state conditions. The differences in these profiles are quite small, but the energy consumptions in the vaporizer are quite different because the sidestream flow rates are different.

The lower graph in Figure 5 presents the control problem in a direct way. The reflux ratio is fixed at 10.96, and the bottoms

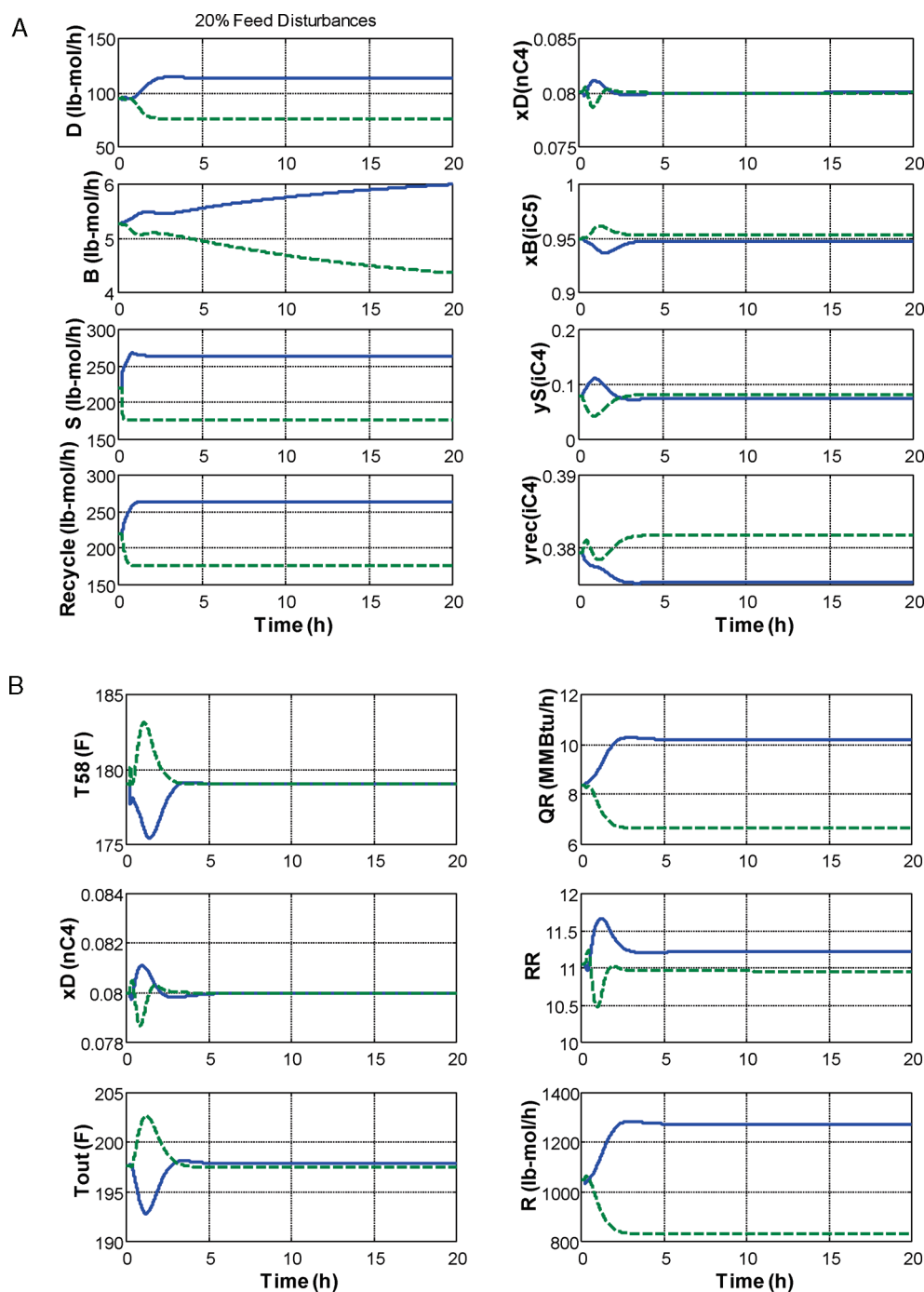


Figure 8. A. 20% feed flow rate disturbances and B. 20% feed flow rate disturbances.

composition is held at 95 mol % iC5 by manipulating bottoms flow rate. The effect of changing sidestream flow rate on the distillate impurity of nC4 is shown in this graph. The curve is not monotonic, which means that the gain between composition and sidestream flow rate is negative for small sidestream flow rates but becomes positive at high stream flow rates. The action of the composition controller would have to be changed depending on the situation.

Since the effect of reflux ratio on distillate purity is monotonic, the control structure selected is one in which reflux ratio is manipulated to control distillate purity and sidestream flow rate is simply ratioed to column feed flow rate.

6.1. Control Structure. Figure 7 shows the plantwide control structure developed for this coupled column/reactor process. Previous work^{1–3} studied the design and control of similar coupled reactor/column systems. Conventional PI controllers are used in all loops. All level loops are proportional with $K_C = 2$ except for the reflux-drum level controller, which uses a gain of 4 since reflux controls reflux-drum level in the high reflux-ratio column.

The temperature profile given in Figure 3 A indicates that the temperature on Stage 58 can be used to maintain bottoms iC5 purity. A distillate composition controller adjusts the reflux ratio. The temperature controller has a 1-min deadtime, and the

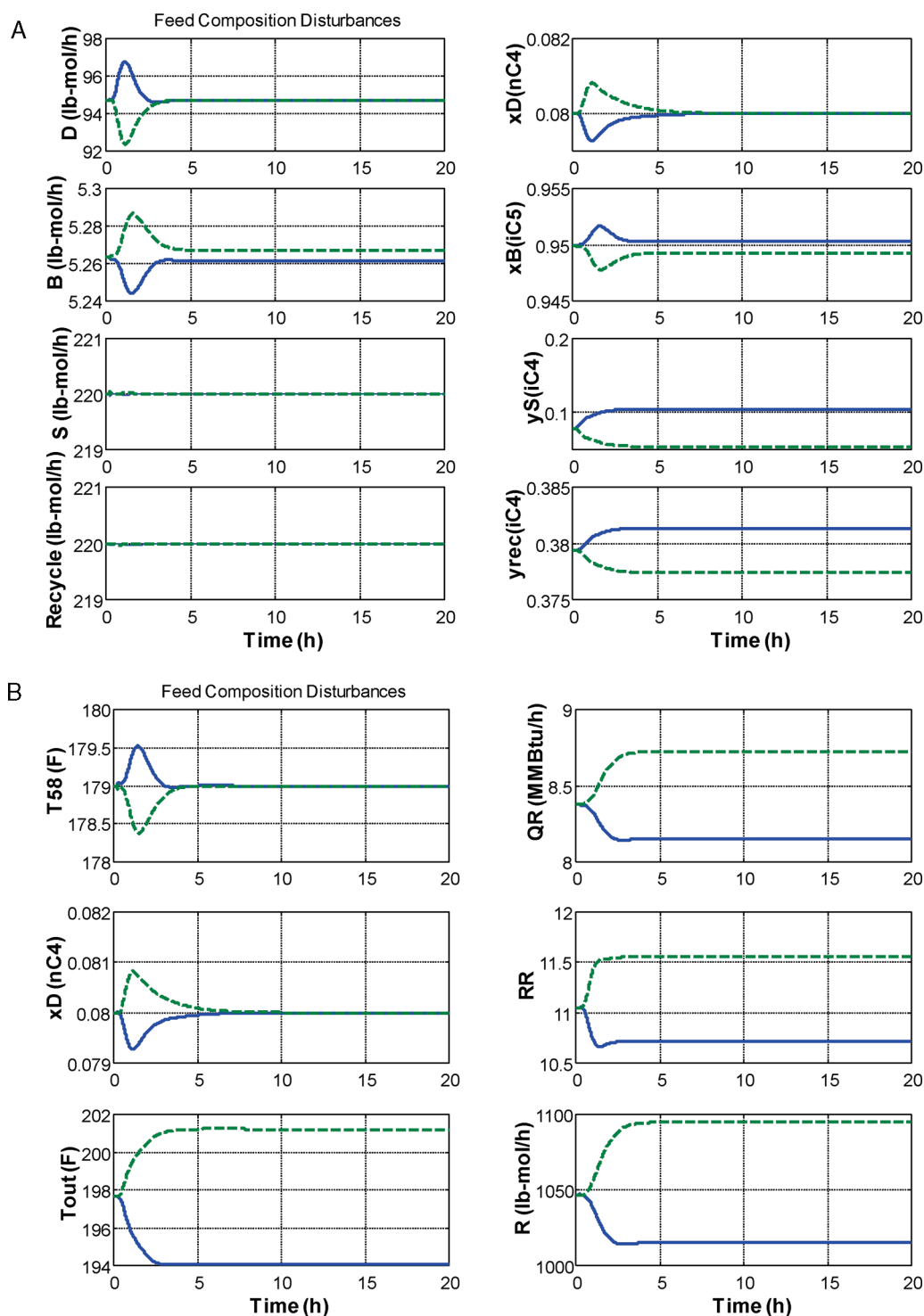


Figure 9. A. Feed iC4 composition disturbances and B. feed iC5 composition disturbances.

composition controller has a 3-min deadtime. These temperature and composition controllers are tuned by using relay-feedback tests to obtain ultimate gains and periods and then applying Tyreus-Luyben tuning rules.

All of the loops are listed below with their controlled and manipulated variables.

- 1 Column feed is flow controlled.
- 2 Sidestream flow rate is flow controlled and ratioed to column feed

- 3 Reflux-drum level is controlled by manipulating reflux.
- 4 Distillate is flow controlled with the set point signal coming from a multiplier whose inputs are the measured reflux flow rate and the desired D/R ratio (reciprocal of the reflux ratio).
- 5 The D/R ratio is manipulated by the distillate composition controller. If the impurity of nC4 increases, then the ratio is decreased (reverse action) to reduce distillate flow rate. The level controller then increases reflux flow rate to lower nC4 impurity.

- 6 Stage 58 temperature is controlled by manipulating reboiler duty.
- 7 Column pressure is controlled by manipulating condenser duty.
- 8 Sidestream drum pressure is controlled by manipulating sidestream condenser duty.
- 9 Sidestream drum level is controlled by manipulating the feed to the vaporizer.
- 10 Vaporizer level is controlled by manipulating vaporizer heat input.
- 11 Reactor pressure is controlled by the valve in the reactor effluent line back to the column.

6.2. Dynamic Performance Results. Several large disturbances are made to test the ability of the proposed control structure. These disturbances include column feed flow rate and feed composition (having more or less iC4 in the feed stream).

A. Throughput Disturbances. Figure 8 gives results for 20% changes in the set point of column feed flow controller. The solid lines are 20% increases; the dashed lines are 20% decreases. The disturbances are made at 0.2 h.

Stable regulatory control is achieved. Most of the transients die out in less than 3 h, but the bottoms flow takes a long time to arrive at its new steady state because it is a very small fraction of the feed. Distillate and bottoms compositions are maintained quite close to their specifications.

Higher feed flow rates require higher reflux ratios to maintain distillate composition in this reactor/column system. In a simple distillation column, the reflux ratio would not change with throughput. In the reactor/column system, the residence time in the reactor changes with throughput, so reactor conditions must change to achieve the required production of iC4 and these affect conditions in the column.

B. Feed Composition Disturbances. Figure 9 gives results for changes in the column feed compositions. The design values are 20 mol % iC4 and 74 mol % nC4. The solid lines show results for an increase in iC4 to 25 mol % and a corresponding reduction in nC4. The dashed lines show results for a decrease in iC4 to 15 mol % and a corresponding increase in nC4.

Having more iC4 in the feed means less iC4 needs to be produced in the reactor. Therefore the change in composition between the inlet (y_S) and the outlet (y_{rec}) becomes smaller, as shown in the lower two right graphs in Figure 9A. Since there is less reaction, the adiabatic reactor temperature changes is smaller, which drops the exit temperature T_{out} (lower left graph in Figure 9B).

The load on the column separation is also reduced, so a lower reflux ratio is required to achieve the desired distillate purity. There is a corresponding reduction in reboiler duty.

Both products are maintained close to their specified compositions. The proposed control structure is demonstrated to provide effective stable control in the face of quite large disturbances.

7. CONCLUSION

A control structure for the coupled reactor/column isomerization process has been developed and tested. It handles large disturbances and maintains both bottoms and distillate products close to the specified purities.

The major control structure question is to select between the two alternative manipulated variables of reflux ratio and sidestream flow rate. Reflux ratio is used because it avoids possible

control problems that could arise due to multiple steady states with the sidestream flow manipulation structure.

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